```
    var senderKey = deriveKey(hkdfKey, senderId, null,
algorithm);    #E
    var recipientKey = deriveKey(hkdfKey, recipientId, null,
algorithm);    #E
    var commonIv = deriveCommonIV(hkdfKey, null, algorithm,
13);    #E
}
```

#A The default algorithm used by OSCORE
#B The Master Key and Master Salt from the OSCORE test case
#C Derive the HKDF master key
#D The Sender ID is an empty byte array, and the Recipient ID is a single 1 byte
#E Derive the keys and Common IV

GENERATING NONCES

The Common IV is not used directly to encrypt data because it is a fixed value, so would immediately result in nonce reuse vulnerabilities. Instead the nonce is derived from a combination of the Common IV, the sequence number (called the Partial IV), and the ID of the sender as shown in listing 13.11. First the sequence number is checked to make sure it fits in 5 bytes, and the Sender ID is checked to ensure it will fit in the remainder of the IV. This puts significant constraints on the maximum size of the Sender ID. A packed binary array is generated consisting of the following items, in order:

- The length of the Sender ID as a single byte.
- The sender ID itself, left-padded with zero bytes until it is 6 bytes less than the total IV length.